| | | |
|---|---|---|
| 11(a)(i) | $E = mc^2$ | **C1** |
| | $= 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$<br><br>$= 8.2 \times 10^{-14}$ J | **A1** |
| 11(a)(ii) | $p = h / \lambda$ **and** $E = hc / \lambda$<br>**or**<br>$E = pc$ | **C1** |
| | $p = (8.2 \times 10^{-14}) / (3.0 \times 10^8)$<br><br>$= 2.7 \times 10^{-22}$ N s | **A1** |
| 11(b) | total momentum (before and after interaction) is zero<br>**or**<br>momentum must be conserved (in the interaction)<br>**or**<br>momentum of the photons must be equal and opposite | **B1** |
| | (photons emitted in) opposite directions | **B1** |